

MATH 3312, SPRING 2026: HOMEWORK 3

- (1) Determine if the following statements are true or false. Give a line or two in justification.
- (a) There exist distinct irreducible polynomials $f(x), g(x), h(x) \in \mathbb{F}_p[x]$ with leading coefficient 1 such that

$$f(x)g(x) = h(x)^2.$$
 - (b) A non-zero polynomial $f(x) \in k[x]$ (k a field) is irreducible if and only if the quotient ring $k[x]/(f(x))$ is an integral domain.
 - (c) If $f(x) \in \mathbb{Z}[x]$ is irreducible, then $\mathbb{Z}[x]/(f(x))$ is a field.
 - (d) A finite field k with 13^2 elements contains an element of (multiplicative) order 5.
 - (e) The quotient ring $\mathbb{F}_{19}[x]/(x^2 - x + 1)$ is a field.

- (2) Suppose that $k \leq L$ is a field extension and that $\alpha \in L$. Show that exactly one of the following is true:
- (a) The evaluation homomorphism $\text{ev}_\alpha : k[x] \rightarrow L$ is injective;
 - (b) There exists a unique irreducible polynomial $m_\alpha(x) \in k[x]$ with leading coefficient 1 and an isomorphism of field extensions of k

$$k[x]/(m_\alpha(x)) \xrightarrow{\cong} \text{im } \text{ev}_\alpha \leq L.$$

*In case (2), we will say that α is **algebraic over k** . The polynomial $m_\alpha(x)$ is called the **minimal polynomial of α (with respect to k)**. If we want to emphasize the role of k , we will also denote it by $m_{\alpha,k}(x)$.*

- (3) Show that the element $\alpha = \sqrt[3]{2} + 2 \in \mathbb{R}$ is algebraic over $\mathbb{Q}$, and compute both $m_{\alpha,\mathbb{Q}}(x)$ and $m_{\alpha,\mathbb{R}}(x)$.
- (4) Suppose that $k_1 \leq k_2$ is a field extension with k_1 and k_2 both finite.
- (a) Show that every element $\alpha \in k_2$ is algebraic over k_1 .
 - (b) If $|k_1| = p^{d_1}$ and $|k_2| = p^{d_2}$, show that $d_1 \mid d_2$.
- (5) Let $f(x) \in \mathbb{F}_p[x]$ be an irreducible polynomial of degree r . Show that the following are equivalent:
- (a) $r \mid d$.
 - (b) $f(x)$ is an irreducible factor of $x^{p^d} - x$ in $\mathbb{F}_p[x]$.
 - (c) There exists a field k of order p^d with $r \mid d$ such that $f(x)$ admits a zero in k .
- (6) Fix $d \geq 1$. Let $\text{Irr}_d(\mathbb{F}_p)$ be the set of irreducible polynomials $f(x) \in \mathbb{F}_p$ with leading coefficient 1 such that $\deg f(x)$ divides d . Show that we have

$$x^{p^d} - x = \prod_{f(x) \in \text{Irr}_d(\mathbb{F}_p)} f(x) \in \mathbb{F}_p[x]$$

Hint: The key is to know that there are no repeated factors. How would you check if a polynomial has repeated irreducible factors?

- (7) Use the previous problem to show that there exist irreducible polynomials in $\mathbb{F}_p[x]$ of every possible degree and hence establish the existence of finite fields of all possible prime power orders.

You can actually use Problem 6 to establish a formula for the number of irreducible polynomials of any degree.